



An augmented reality-based wearable system for handheld-free and intuitive robot programming

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Abstract

Industrial robots have been programmed to optimize mass production by executing predefined trajectories with their high speed, precision, and repeatability over long production cycles. However, the rise of Industry 4.0 and the increasing demand for diverse product varieties in small quantities have sparked the need for new programming methodologies that enable rapid reprogramming and short production cycles. Recently, augmented reality (AR)-based robot programming has gained attention for overlaying virtual images into real working environments, making programming intuitive and fast. However, the discomfort from prolonged use of AR-handheld devices, as well as the accuracy and latency issues associated with vision-based hand tracking, limit their usability, and impede effective human–robot cooperation. To overcome these limitations, this paper introduces an augmented reality (AR)-based wearable robot programming system using a haptic glove, allowing users to intuitively program the robot in a handheld-free and wearable manner. The use of a haptic glove enables accurate measurements of finger joint angles and handheld-free haptic feedback, eliminating discomfort and occlusion problem. The system leverages measured hand posture and finger joint angles, combined with voice commands, to enable the simultaneous input of various types of commands for intuitive programming. A practical human–robot collaboration scenario demonstration with a 6-degrees of freedom (DOF) hydraulic manipulator to disassemble a ladder for repair shows the programming flexibility of our system. The proposed system addresses the evolving trends of the manufacturing industry by enhancing programming flexibility and intuitiveness, as well as promoting seamless cooperation between humans and robots.

Keywords Augmented reality · Robot programming · Wearable interface · Haptic feedback · Human–robot collaboration

Introduction

Industrial robots have revolutionized manufacturing by significantly enhancing productivity and quality with their remarkable speed, payload capacity, repeatability, and accuracy. Previously, industrial robots performed tasks by repeating predefined trajectories over long cycles with minimal

modifications, focusing on mass production and maximizing efficiency. (Foumani et al., 2014) introduced an approach to maximize production rates by scheduling multifunction robots (MFRs) that can perform additional tasks during transit between machines. Furthermore, (Szczepanski et al., 2022) showed how the artificial bee colony algorithm can be used to improve the scheduling of robotic arms for palletizing tasks, enhancing the speed and accuracy of manufacturing processes.

But, with the advent of Industry 4.0 and the rapid growth of start-ups, manufacturing is now facing a new trend of producing a diverse variety of items in small quantities (Cohen et al., 2019). This emerging trend demands new programming methodologies that enable quick robot reprogramming, support short production cycles, offer flexibility, and are user-friendly for non-experts. Additionally, this trend emphasizes the growing significance of human–robot collaboration, which combines human cognitive abilities and flexibility

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with the robot's high speed, and repeatability (Faccio & Cohen, 2024). This collaborative approach aims to attain elevated levels of productivity, flexibility, and adaptability in the manufacturing industry.

However, conventional robot programming techniques, such as online and offline methods, are time-consuming and require significant expertise (Liu et al., 2010; Pan et al., 2012; Ragaglia et al., 2016; Zheng et al., 2022). The online programming method uses a teaching pendant or lead-through approach to program or modify trajectories (Pan et al., 2012). Through a teaching pendant, such as a joystick, users can program precise trajectories by specifying the desired position, orientation, and gripper stroke of grippers. On the other hand, the lead-through method, also known as kinesthetic teaching, involves the operator physically moving the robot to demonstrate the desired motions (Ragaglia et al., 2016). A major drawback of these methods is the need to input position, orientation, and gripper stroke separately, making the programming process time-consuming. Additionally, they typically lack support for visual trajectory representation, thereby reducing their overall intuitiveness.

In contrast, offline programming eliminates the need for the physical presence of robots by creating their virtual representation within a simulated workplace without interrupting task operations (Liu et al., 2010; Zheng et al., 2022). This allows for advanced path planning, optimization, and collision avoidance techniques, which lessen the path designers' workload and enhance performance. However, developing and modifying virtual environments to meet high simulation fidelity and adapting to changed working environments require extensive time and expertise.

In response to the limitations of traditional programming techniques, an advanced method called Programming by Demonstration (PbD) has been widely used in manufacturing (Hu & Xiong, 2019; Pardowitz et al., 2007; Zhou et al., 2020). PbD enables operators to teach robots desired tasks by physically demonstrating the motions themselves instead of manually specifying trajectories. Therefore, programming time and complexity can be reduced, and flexible production and easy robot adaptation become possible. In recent advancements, the implementation of data gloves has significantly enhanced the capture and interpretation of human manipulation, allowing for more accurate and detailed inputs in PbD systems (Fresnillo et al., 2023). However, its performance, particularly accuracy, heavily depends on the quality of demonstrations, requiring considerable expertise and labor-intensive training data.

Augmented reality (AR)-based robot programming has gained attention for its ability to enhance intuitiveness and significantly simplify robot programming, addressing the problems of previous programming methods (Aivaliotis et al., 2023; Akhmetov & Varol, 2023; Azuma, 1997; Chu & Weng, 2024; Gattullo et al., 2019; Lambrecht et al., 2021;

Li et al., 2023; Liu et al., 2024; Ong et al., 2010, 2020; Yang et al., 2024). For instance, De Boer et al. (2023) showed that 3D teleoperation using AR-based visual feedback yielded smoother and more precise control in a robotic manipulation task compared to traditional 2D interfaces. Furthermore, AR applications for end-user robot programming have been developed to empower non-experts to engage in programming tasks intuitively (Kapinus et al., 2024). Systems such as PProgramAR, which combines AR with Trigger-Action Programming (TAP), allow users to define reactive behaviors without needing prior expertise, supporting a range of collaborative and assistive tasks (Ikeda & Szafir, 2024).

This approach overlays virtual objects and information, including robots and desired trajectories, onto the real-world environment and intuitively visualizes a preview of robot movements. Therefore, AR enhances robot programming by providing real-time adjustments, intuitive visual feedback, and predictive collision visualization, making interfaces more user-friendly and adaptable to dynamic environments. In order to facilitate a sense of depth and spatial awareness, and to enhance field-of-view (FOV) and intuitiveness, 3D head-mounted displays (HMDs) are commonly used, instead of relying on 2D vision like projecting AR image onto the surface (Antonelli & Astanin, 2016; Doshi et al., 2017; Leutert & Schilling, 2022) or holding smartphones or tablets (Fang et al., 2014; Frank et al., 2016; Stadler et al., 2016).

Handheld devices like AR/VR controllers or vision-based hand-tracking methods have been employed to input robot paths in AR-based robot programming (Fang et al., 2014; Ong et al., 2020; Ostanin & Klimchik, 2018; Pai et al., 2014; Quintero et al., 2018; Rosen et al., 2019). Handheld devices are used to input commands and interact with virtual content in AR environments through their buttons, joysticks, and vibrators (Fang et al., 2014; Ong et al., 2020; Pai et al., 2014). They are typically designed to be compact, lightweight, and portable, and feature built-in sensors such as accelerometers, gyroscopes, and GPS to enable motion sensing for convenient on-the-go interaction. Moreover, haptic feedback from handheld devices, such as vibrations or tactile sensations, enhances users' touch perception and manipulation of virtual objects (Burke et al., 2006). Haptic feedback in robot programming not only improves intuitiveness, but also informs users of potential collisions, enabling safer and more efficient programming. However, these devices have limitations, including the challenge of simultaneously pressing multiple buttons and the potential for fatigue and discomfort with prolonged handling (Makhataeva & Varol, 2020).

Whereas vision-based hand or finger tracking techniques enable users to interact with the virtual environment through natural hand gestures that mimic real-world actions without using handheld devices (Bordoni & Tang, 2023; Ostanin & Klimchik, 2018; Quintero et al., 2018; Rosen et al.,

2019). These techniques use RGB, depth, or infrared cameras, along with computer vision algorithms, to detect and track users' hand or finger movements in real-time. They also offer the capability to simultaneously input various types of commands easily. However, accuracy, latency, and occlusion issues can occur due to lighting conditions, complex hand poses, and obstruction from view, which can limit environmental conditions. Furthermore, vision-based hand tracking provides limited haptic feedback to the hand or fingers, thereby restricting intuitiveness and touch or error perception (Liau & Ryu, 2024). These limitations not only restrict usability but also hinder effective cooperation between humans and robots.

Thus, in this paper, an AR-based wearable robot-programming system using a haptic glove is proposed to solve the mentioned problems, specifically enhancing the intuitiveness and flexibility of robot arm programming, as shown in Fig. 1. The contributions of the proposed system are as follows:

- 1) The use of a haptic glove allows for measuring finger joint angles without requiring the hand to be within the FOV of the AR HMD, a relatively narrow FOV in a 4:3 aspect ratio with horizontal, vertical, and diagonal FOVs of 40, 30, and 50 degrees, respectively. Furthermore, it addresses the issue of recognition errors caused by fingers being obscured by the back of the hand or the opposite arm, even within the FOV of the AR HMD. This capability not only overcomes these challenges but also offers flexibility and high adaptability across various tasks.
- 2) By utilizing measured hand postures and finger joint angles, along with voice commands, it becomes possible to simultaneously input various command types including a desired position, orientation, and gripper stroke through natural hand gestures that closely mimic real-world actions. Furthermore, this approach eliminates the need to repeatedly pick up and put down handheld devices during cooperative tasks.
- 3) The use of haptic gloves provides handheld-free haptic feedback, eliminating the discomfort of prolonged use and enhancing the tactile experience during programming. Fingertip vibrations are provided to convey crucial information regarding collisions, out-of-range movements, and the gripper's closed state, thereby enhancing the intuitiveness of trajectory programming.

Within these advantages, our approach effectively addresses the evolving trends and demands of the field, including short production cycles and flexibility, while also offering improved intuitiveness for non-experts and promoting seamless cooperation between humans and robots.

The remainder of the paper is organized as follows. In Sect. “[System design and implementation](#)”, the proposed programming method using wearable devices is explained with details of the programming steps. In Sect. “[Applications](#)”, a practical human–robot collaboration scenario shows the effectiveness of the proposed system. Finally, in Sect. “[Conclusion and outlook](#)”, the paper is summarized, and future works are outlined.

System design and implementation

System overview

The AR-based wearable robot programming system comprises a wearable haptic interface, a hydraulic robot arm, and an AR programming algorithm, as depicted in Fig. 1. This system allows users to conveniently program the robot in a handheld-free manner while viewing a virtual image superimposed on the real work environment and receiving critical information through haptic feedback. Furthermore, by utilizing measured hand postures and finger joint angles, along with voice commands, it becomes possible to simultaneously input various command types, enabling intuitive (re)programming.

The wearable haptic interface includes the AR head-mounted display (HMD), the haptic glove, and the remote AR/VR controller. Magic Leap One (Magic Leap One, 2018) is used as the AR HMD; it weighs only 316 g and contains a built-in microphone for voice control. The HMD overlays virtual objects, such as robot trajectories, end-effector positions, and collision boundaries, directly onto the physical robot and its surrounding workspace. This augmented view allows users to see a real-time visualization of the robot's movements and intended actions, seamlessly integrated into the real environment. By providing immediate feedback on spatial alignment and potential obstacles, the HMD enables users to preemptively detect and correct errors, reducing the need for trial-and-error adjustments and enhancing the efficiency and accuracy of programming. While this relatively narrow FOV may constrain the user's ability to visualize virtual objects overlaid in the workspace, the system mitigates this limitation through the integration of the haptic glove. The glove enables users to interact with the robot through natural hand gestures and tactile feedback, reducing the reliance on constant visual input.

The IBM Watson voice recognition system (IBM Watson, 2016) converts spoken words into text strings in real time. Mollisen HAND (Mollisen Hand Vib+, 2019) is used as the haptic glove, which has embedded soft sensors that accurately measure two joint angles per finger and a total of 10 joint angles per hand. But in this work, only the joint angles of the thumb, index, and middle fingers are used, as shown



Fig. 1 Overview of the entire system. The user can program the robot trajectory by wearing a wearable haptic glove without handheld devices. Hand and finger tracking enables rapid and intuitive programming,

while operators receive fingertip vibration feedbacks on crucial information such as gripper stroke closure, out-of-range, and collisions

in Fig. 1. The haptic glove provides vibration feedback to the tips of the thumb, index, and middle fingers. The remote AR/VR controller is securely attached to the back of the haptic glove, allowing tracking of hand position and orientation.

Our research group developed the industrial hydraulic robot arm, which features 6 degrees of freedom (DOF) in its serial joints, a 1-DOF gripper, a maximum arm length of 2.8 m, and can handle payloads up to 80 kg. The robot was operated in real time by LabVIEW software and a cRIO controller, which includes an inverse kinematics algorithm for numerically calculating joint angles for a given position and orientation in the target trajectory. The AR programming algorithm, running within the Unity software, seamlessly integrates the wearable haptic interface, real robot, AR-simulated robot, and the programmed trajectory.

Referencing the AR-simulated robot

During robot programming in the AR environment, the AR-simulated robot needs to be superimposed onto the real robot arm with the same local coordination. To achieve this referencing process in our system, a reference image is attached to the base frame of the real robot (Fig. 2a), which is already uploaded to the software. When the user wears the AR headset and focuses on the reference image, the image tracking functionality of Magic Leap's Unity software development kit (SDK) automatically detects the image. The system calculates the image's position and orientation, and then superimposes the pre-loaded 3D model of the robot within Unity to align with these coordinates. As a result, both the real robot and the AR-simulated robot are aligned accurately (Fig. 2b).

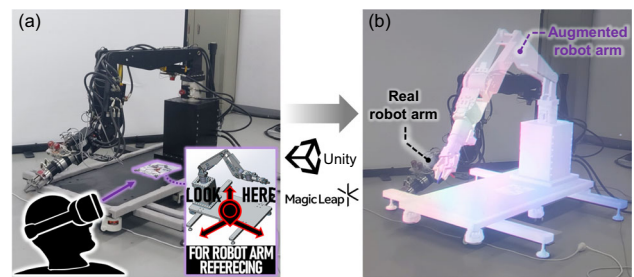


Fig. 2 Robot referencing. **a** A referencing image on the robot arm. **b** An augmented robot overlaid on the real robot

Specifying a position, orientation, and gripper stroke

Users can specify the points that the robot gripper should pass through, including position, orientation, and gripper stroke, via natural hand and finger gestures measured by the wearable haptic interface. Extending the thumb activates tracking mode, allowing the user to move or rotate their hand to specify the gripper's position and orientation. If the thumb angle exceeds a certain threshold, the gripper's position and orientation synchronize with the user's hand movements, as shown in Fig. 3a. This intuitive 'drag and drop' method overcomes the limited range of motion (ROM) of the user's hand, making it particularly useful for programming large robots. The gripper stroke is controlled by flexing the index finger, with the degree of flexion directly mapping to the gripper's opening or closing, as depicted in Fig. 3b. These gestures were specifically chosen for their alignment with natural human actions, simplifying interaction and reducing cognitive effort compared to traditional button-based methods.

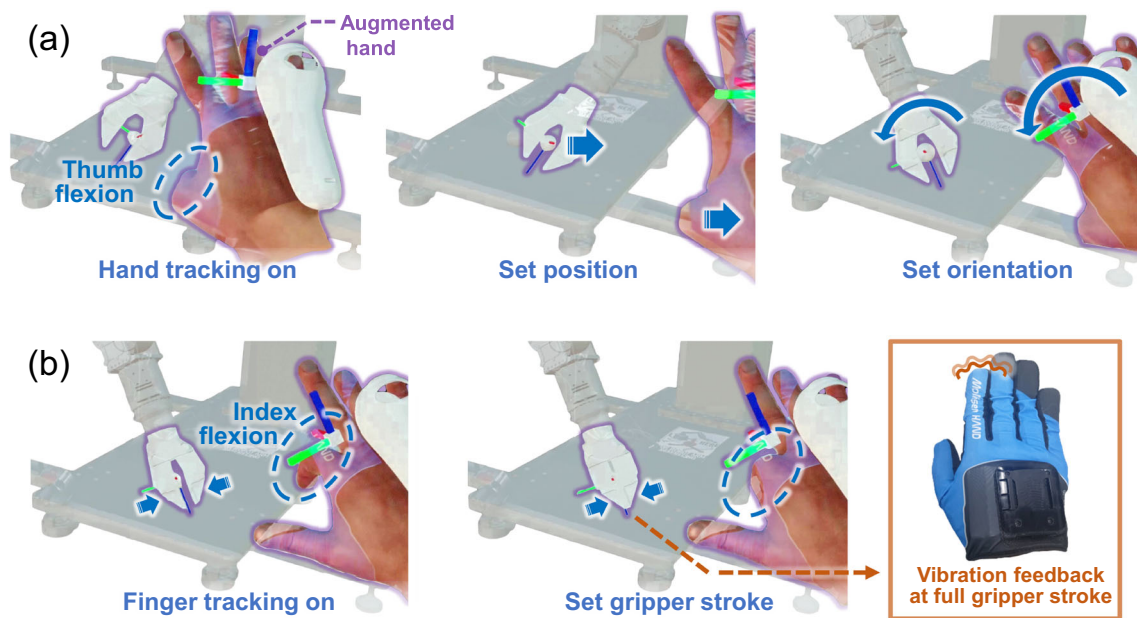


Fig. 3 Specifying **a** desired position, orientation, and **b** gripper stroke using the wearable haptic interface

To enhance usability, the system provides tactile feedback through vibrations at the index fingertip when the gripper is fully closed, addressing the difficulty of determining closure using only visual feedback in the AR HMD. This feature is especially important when manipulating thin plates or wires. By combining thumb and index finger gestures with tactile feedback, users can simultaneously input multiple types of commands, including position, orientation, and gripper stroke, in a seamless and ergonomic manner. These intuitive gestures significantly improve programming flexibility and efficiency, particularly during collaborative tasks.

Real-time haptic feedback for ROM and collision detection

Providing haptic feedback to the user can effectively prevent errors caused by user mistakes or changes in the environment during programming, enhancing error recognition and prevention. If the virtual end-effector exceeds the range of motion (ROM) or collides with scanned obstacles, real-time vibration feedback is sent to the fingertips as a warning, as illustrated in Fig. 4. In our AR application, we utilize the meshing functionality provided by Magic Leap's Unity SDK to transform the physical environment into a detailed digital mesh in real-time. This feature plays a critical role in enabling precise collision detection and enhancing environmental awareness in augmented reality scenarios. The system employs the embedded depth camera in the AR glasses to continuously scan and capture the three-dimensional spatial features of the workspace as users traverse their environment. The captured data is processed to generate a triangular mesh

surface that accurately defines the physical boundaries of both static and movable objects. The entire meshing process, from initial scan to mesh generation, is completed within approximately three minutes, ensuring a fast and efficient setup for AR-based operations. Also, the HMD display indicates the type of error with the label 'Out' or 'Detected' for out-of-range or collision, respectively.

Programming process

The system architecture for the overall trajectory programming is summarized as follows (Fig. 5). Firstly, the user interacts with the AR-augmented robot and the environment using the trajectory programming algorithm, which is implemented within the Unity program. The wearable haptic device allows users to give commands via voice, finger joint angles, hand position, and orientation, and receive 3D visual and fingertip vibration feedback during this process. Upon completion in Unity, the trajectory, programmed in Cartesian coordinates, is sent via TCP to the inverse kinematics and robot control algorithms running in LabVIEW. Eventually, the trajectory is converted into joint space using the inverse kinematics algorithm, and the robot control algorithm takes over and controls the robot according to the specified trajectory.

Our system's error-checking algorithm ensures the reliability and safety of the system by verifying the functionality of all devices, software, and communications. Before operation, the algorithm systematically checks hardware components, including the AR glasses, haptic glove, and robotic controller, to confirm proper connections and functionality.

Fig. 4 Handheld-free vibration feedback to detect **a** out-of-range motion and **b** collision with obstacles during the robot programming process

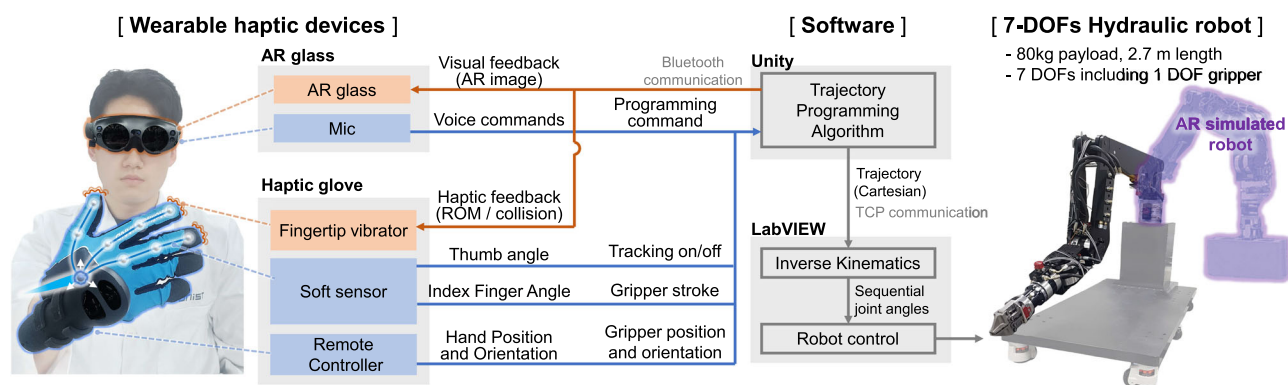
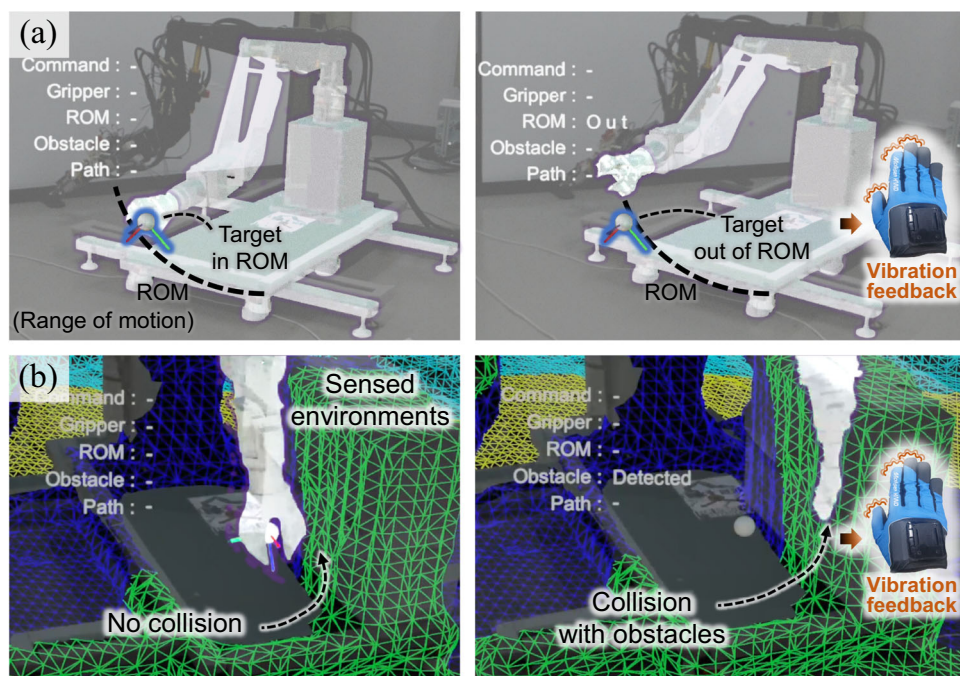


Fig. 5 System architecture for haptic feedback and AR-based control of a 7-DOF hydraulic robot via wearable haptic devices

For software, it validates the integrity of the AR programming environment, trajectory generation, and inverse kinematics module to identify potential inconsistencies or errors. The system leverages a straightforward serial connection configuration, where devices are directly linked to Unity software and communicate with LabVIEW using TCP, simplifying integration and facilitating error identification. Additionally, Bluetooth connectivity and TCP communications are automatically verified to ensure seamless device interaction.

During operation, the algorithm continuously monitors data flow between devices to detect anomalies such as unexpected delays, dropped packets, or corrupted data. Sensor readings from the haptic glove and robotic arm are validated to identify sudden peaks or unexpected jumps, which may indicate sensor malfunctions or environmental interference. In the event of such data errors, the system either corrects the values or halts the robot and returns it to a safe position.

Critical safety measures include the ability to manually stop the hydraulic manipulator via an emergency button during operation.

The trajectory programming algorithm involves several steps, as illustrated in the flowchart shown in Fig. 6. To begin, the user performs robot referencing. The user specifies a desired point, including position, orientation, and stroke of the gripper, as illustrated in Fig. 3. During this process, if any errors occur, such as exceeding the ROM or colliding with obstacles, vibration feedback is delivered to the fingertip as shown in Fig. 4, prompting the user to modify the point. Assuming no errors occur at the specified point, when the user enters the voice command 'here', the virtual end-effector is visually highlighted in a light green color, indicating the location of the specified point, as shown in Fig. 7a. Similarly, the user iteratively specifies desired points, as depicted in Fig. 7b. Entering the voice command 'okay' generates interpolation

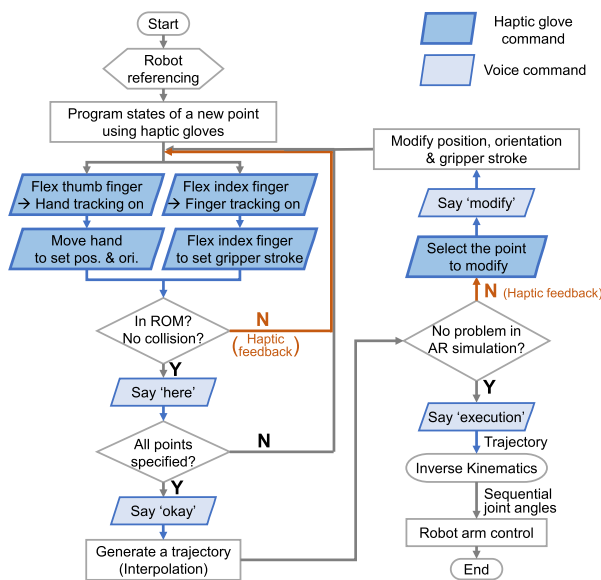


Fig. 6 Flowchart of the trajectory programming algorithm. The color red represents feedback, while the color blue represents commands transmitted through wearable devices (Color figure online)

points using a cubic spline between the specified points, as shown in Fig. 7c. As shown in Fig. 7d, when the interpolated points collide with an obstacle mapped by the AR HMD, fingertip vibrations are transmitted, and the HMD display indicates the collision with a 'Detected' label. If the user identifies a point requiring correction—whether it is outside the workspace, colliding with obstacles, or inaccurately specified due to an incorrect finger joint angle or hand position—they can issue the voice command "modify" to activate the point

for adjustment. Upon activation, the system highlights the problematic point with a translucent red sphere, making it easily distinguishable in the augmented reality interface. This visual indicator, combined with tactile feedback through the haptic glove, guides the user to precisely locate and correct the point's position, orientation, or gripper stroke using the same intuitive gestures as in the initial programming process. The point can be modified in the same process, and if there are no problems with all points later, the 'detected' label and fingertip vibration disappear as shown in Fig. 7e. When the voice command 'execution' is entered, the trajectory is transferred to the LabVIEW program over TCP communication, and the robot control algorithm operates the robot along the sequential joint angles calculated by the inverse kinematics. Subsequently, the robot safely transports the bricks while avoiding the obstacle, as demonstrated in Fig. 7f–h.

Applications

A human–robot collaboration scenario was demonstrated to show the effectiveness of the suggested AR-based wearable haptic programming system. The scenario assumed that the user and the robot collaborated to disassemble a ladder for repair while the user programmed the robot's trajectory. As shown in Fig. 8a–c, the user programmed the first trajectory, whereby the robot lifted a heavy ladder and held it horizontally while the user removed bolts on the right side of the ladder with an electric screwdriver. Once the first collaborative task was completed, the user programmed the second trajectory, whereby the robot rotated the ladder and held it while the user removed bolts on the opposite side of the

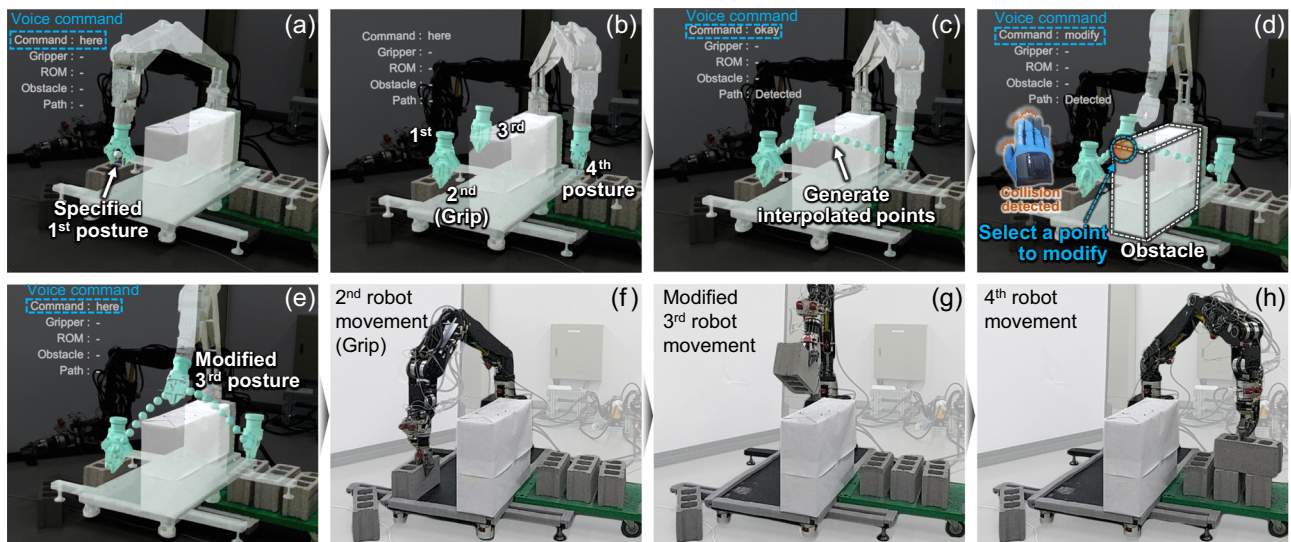


Fig. 7 Trajectory programming process and robot operation. **a, b** Specifying points. **c** Generating an interpolation line. **d, e** Selecting and reprogramming a point to be modified. **f–h** Actual robot movement along the programmed trajectory

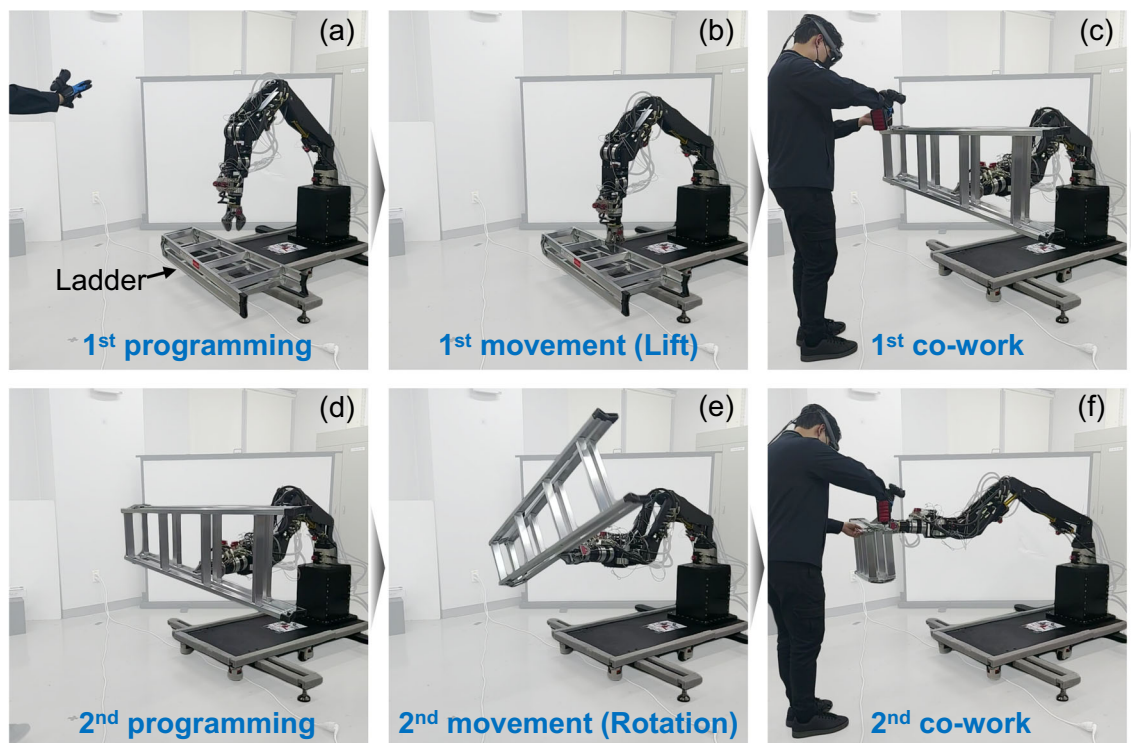


Fig. 8 Human–robot collaboration scenario to disassemble a ladder for repair. **a–c** First programming: robot movement and co-work. **d–f** Second programming: robot movement and co-work

ladder, as shown in Fig. 8d–f. Because our system offers a handheld-free programming solution, it allows a fast and seamless transition between programming and working or collaborating with robots. This feature makes the manufacturing process more efficient and flexible, as the average time required to complete the scenario is less than two minutes.

Conclusion and outlook

This paper presents an AR-based wearable robot programming system using a haptic glove to address the limitations of traditional robot programming methods. The system included the AR head-mounted display (HMD), haptic glove, and remote AR/VR controller. The HMD overlays virtual objects, such as robot trajectories, end-effector positions, and collision boundaries, directly onto the real-world environment, enabling intuitive and real-time visualization. Meanwhile, the haptic glove provides precise measurements of hand and finger gestures and delivers handheld-free haptic feedback, enhancing interaction and usability. The system was integrated with the industrial hydraulic robot, and the programming process was facilitated through Unity and LabVIEW software.

Our proposed system effectively addressed the challenges posed by short production cycles, flexibility requirements, and the need for improved intuitiveness in robot programming. The implementation of a haptic glove overcomes

discomfort and occlusion issues associated with handheld controllers or vision-based tracking systems. While the AR HMD has a relatively narrow field of view (40° horizontally, 30° vertically, and 50° diagonally), the system compensates for this limitation by integrating tactile feedback and gesture-based interactions. In addition, it enables robust measurement of finger joint angles, allowing users to interact naturally with the robotic interface. By leveraging gestures that mimic real-world actions, such as extending the thumb for tracking and flexing the index finger to control the gripper stroke, the system aligns closely with human motor skills, simplifying interaction and reducing cognitive effort.

Furthermore, handheld-free haptic feedback enhances programming intuitiveness by providing real-time tactile cues, such as vibrations to signal collisions, out-of-range movements, or when the gripper is fully closed. This immediate feedback improves error recognition and facilitates a more immersive and informed programming experience. These features not only make programming more intuitive and user-friendly but also reduce physical strain, allowing for extended use during collaborative tasks. As a result, the system supports seamless cooperation between humans and robots, particularly in dynamic manufacturing environments, by improving flexibility. Programming flexibility was shown in the practical human–robot collaboration scenario by demonstrating the ease of performing other tasks or collaborating with robots while programming tasks.

In conclusion, our AR-based wearable robot programming system with a haptic glove provides a promising approach to intuitive and flexible robot programming in manufacturing settings. The system's advantages, including enhanced intuitiveness, simultaneous input of commands, handheld-free haptic feedback, and improved collaboration between humans and robots, make it well-suited for addressing the evolving needs and trends in the industry. This study lays the groundwork for enabling even non-experts to program robot trajectories effortlessly and accurately using haptic gloves. By integrating intuitive AR overlays and wearable devices, it addresses key challenges like expertise requirements and operational constraints. This approach paves the way for broader adoption in both industrial and non-industrial settings. Future work will concentrate on enhancing the system's performance by incorporating advanced artificial intelligence-based real-time gesture recognition algorithms. For example, the movement speed ratio between the hand and the gripper can be adjusted by using the non-dominant hand to balance between rapid and precise programming. In forthcoming studies, we will conduct comparisons between our programming method and traditional programming methods to clearly demonstrate the effectiveness of the new system.

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Declarations

Conflicts of interest The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Research involving human and animal participants Not applicable as this research did not involve any human participants or animals.

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